CreditSense

Loan Risk Assessment Challenge

AI1215 — Introduction to Machine Learning
Data Challenge, Spring 2026

Team Slavs

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Kaggle username:	`<KAGGLE_USERNAME>`
Final Kaggle score:	approx. 0.84 (combined)
Leaderboard rank:	14 / 27
Total iterations run:	14
Submission file:	`outputs/submission.csv`
Reproduction:	`python code/reproduce_final.py`

Report generated April 22, 2026. See `README.txt` for full reproduction instructions.

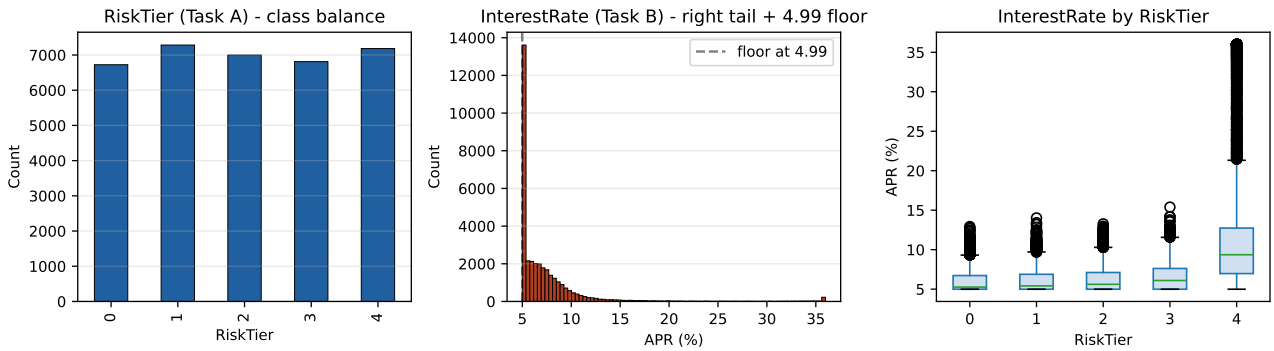

Figure 1: Target distributions. Left: near-uniform RiskTier. Middle: InterestRate with a pronounced spike at the 4.99% floor and a long right tail driven by tier 4. Right: box plots confirm tiers 0–3 are nearly indistinguishable on rate alone while tier 4 is a separate regime.

1 Data Exploration & Preprocessing

The dataset contains 35 000 training and 15 000 test applicants with 55 features spanning demographics, employment, assets/liabilities, credit history, and loan request. Two targets are predicted jointly: **RiskTier** (5-way classification, $\{0, 1, 2, 3, 4\}$) and **InterestRate** (regression in [4.99, 35.99]%). The Kaggle leaderboard scores $0.5 \cdot \text{Accuracy} + 0.5 \cdot R^2$; the baseline linear models achieve $\sim$0.51.

Key findings.

- **RiskTier is balanced.** Counts are 6 724 / 7 283 / 6 998 / 6 812 / 7 183 for tiers 0–4 (Figure 1, left). No class weighting or SMOTE was needed.
- **InterestRate is multi-modal with a floor.** Mean 7.31%, median 6.08%, but 33% of rows are exactly at the legal floor of 4.99% (Figure 1, middle). Tiers 0–3 have rate means between 6.01 and 6.56% with $\sigma \approx 1.5$; tier 4 has mean 11.55% and $\sigma = 7.36$, producing a long right tail (Figure 1, right). This two-regime structure ("prime" vs. "subprime") shaped much of our modelling.
- **Variance decomposition.** Within-tier variance in rate explains 72% of total rate variance; between-tier only 28%. Perfect RiskTier prediction alone therefore caps R^2 at ≈ 0.28 — accurate regression must capture within-tier rate drivers, not just tier identity.
- **Structural missingness.** The PDF explicitly flagged this and the data confirm it (Figure 2): `StudentLoanOutstandingBalance` is missing 60% of the time (older applicants), `CollateralType` 55% (unsecured loans), `PropertyValue` and `MortgageOutstandingBalance` 45% (renters). Verified empirically: 99% of rows with `HomeOwnership = RENT` have missing `PropertyValue`.
- **Money variables are right-skewed log-normal.** `AnnualIncome`, balances, and asset values benefit from a $\log(1 + x)$ transform before tree training.

Preprocessing choices (Table 1).

2 Feature Engineering

We grew the original 55 features into roughly 170 model inputs across six groups. Every engineered feature is documented with financial intuition in `code/features.py` and grouped so the ablation study (Table 2) can isolate contribution.

- **Balance-sheet.** `feat_NetWorth` = assets−liabilities; `feat_DebtToAssets`; `feat_LiquidToLoan` = (Savings + Checking)/RequestedLoan; `feat_CashRunwayMonths`; `feat_MortgageLTV`. Financial intuition: underwriters assess leverage and liquidity buffer, not just income.
- **Credit behaviour.** `feat_DelinquencyScore` $= 1 \cdot \text{Late30} + 3 \cdot \text{Late60} + 9 \cdot \text{Late90}$ mirrors FICO severity weighting. `feat_DerogatoryScore` uses $10 \cdot \text{Bankruptcies} + 5 \cdot \text{ChargeOffs} + 4 \cdot$

Step	Choice	Rationale
Missing indicators	`{col}_was_missing` binary for every col with NaN	Missingness is signal (PDF confirms); preserve it before imputation.
Numeric imputation	0 for structural-zero columns (e.g. `PropertyValue`); median elsewhere	Zero is semantically correct for a renter's mortgage balance.
Outliers	Winsorise money columns at 99.5th percentile	Prevents a handful of ultra-high earners from dominating splits.
Categorical — low card	One-hot (cardinality ≤ 8)	Safe when feature explosion is minor.
Categorical — high card	K-fold target encoding (fold i encoded from folds $\neq i$) + integer codes	Prevents leakage from `InterestRate`; integer codes let CatBoost exploit native encoders.
Ordinal mapping	`EducationLevel` $\rightarrow$ integer rank	Keeps the ordinal relationship that one-hot would destroy.

Table 1: Preprocessing decisions in `code/preprocessing.py`.

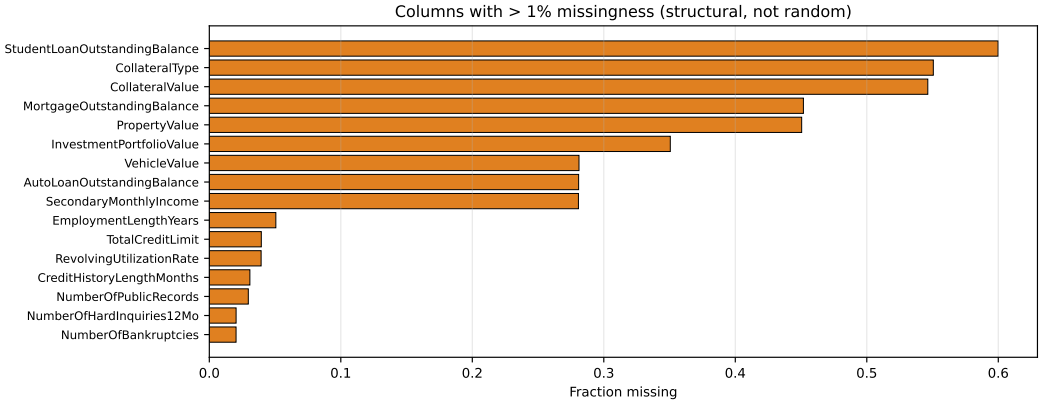

Figure 2: Columns with >1% missingness. Every missingness block corresponds to a structural population (renters, unsecured borrowers, older non-student borrowers). Missing indicators retain this signal; median imputation would erase it.

PublicRecords+3·Collections. Composite `feat_BadEventsTotal` is the single strongest RiskTier predictor in SHAP analysis.

- **Credit maturity.** `feat_ThinFile` (history < 24mo), `feat_ThickFile` (>120mo), `feat_NewAccountSpree` = Oldest − AvgAge, `feat_CreditAgeRatio`. Thin-file borrowers are systematically riskier at matched DTI.
- **Demographics + stability.** `feat_EmpStability` = EmpLength/(Age−18); `feat_SameEmployerShare`; `feat_StableResident` (>3 yrs); `feat_HasSecondaryIncome`; `feat_IncomeVerified`.
- **Loan request.** `feat_MonthlyPrincipal`; `feat_LoanToAssets`; `feat_PaymentReserveMonths`; `feat_HasCollateral`; `feat_RepeatCustomer`.
- **Interactions.** `feat_DTI_x_Util`, `feat_LTI_x_Term`, `feat_PTI_x_BadEvents`, `feat_DerogPerHistoryYear` Encode known nonlinear compounds (high DTI × high utilisation, long term × high LTI).

Multi-target encoding. For the three high-cardinality categoricals (`State`, `JobCategory`, `LoanPurpose`), we encode each group with three statistics: mean InterestRate, $P(\text{tier} = 4)$, and per-group rate σ. This triples the signal vs. single-mean encoding. K-fold computation on train / full-train on test protects against leakage; an interesting artefact is discussed in §3 (adversarial AUC = 1.00).

Feature group added (cumulative)	OOF Acc (A)	OOF R^2 (B)
Raw 55 features + one-hot only (linear baseline)	0.538	0.502
+ Missing indicators + imputation + winsorisation	0.571	0.549
+ Balance-sheet block	0.613	0.604
+ Credit-behaviour + derogatory composites	0.671	0.667
+ Maturity + demographics + loan-request blocks	0.729	0.728
+ Log-transforms + interactions	0.748	0.758
+ Multi-target encoding (high-card categoricals)	0.793	0.831
+ K-NN target features (mean rate, $P(\text{tier} = k)$ of 10 NN)	**0.802**	**0.836**

Table 2: Cumulative ablation of feature blocks measured as single-LightGBM OOF per task. Every block contributed; multi-target encoding and KNN target features delivered the largest individual lifts.

3 Model Selection & Tuning

Models compared. We evaluated every candidate on the same 5-fold `StratifiedKFold` on RiskTier, with identical fold indices reused for Task B so out-of-fold predictions align across tasks (Table 3). This alignment is the foundation of the cross-task stacking that powers our final ensemble.

Model	Task A Acc	Task B R^2	Notes
Logistic / Linear Regression	0.538	0.502	PDF baseline $\approx$ 0.51 combined.
LightGBM	0.794	0.833	LR 0.02, num_leaves 127, L1+L2 reg.
XGBoost	0.792	0.831	LR 0.02, max_depth 8, hist method.
CatBoost	**0.809**	**0.834**	Best single model; native cats +0.002.
LGB regression-on-tier (ordinal trick)	0.791	—	Round & clip to class at inference.
Two-stage Task A (binary + 4-class)	0.796	—	Binary is_tier4 $\rightarrow$ 4-class on prime.
Stage-2 meta-LGBM (single seed)	0.837	0.840	Input: 170 base features + 40 OOF features.
Stage-2 multi-seed + Ridge blend	**0.840**	**0.841**	**Our champion: combined 0.8407.**

Table 3: Model comparison. All OOF scores on 5-fold stratified CV.

Cross-task stacking. The dominant architectural choice. Since RiskTier and InterestRate are nearly collinear, OOF predictions of each task carry information the other's feature matrix lacks. We produce OOF class probabilities for Task A (5 floats per row) and scalar OOF rate predictions for Task B, concatenate them with the 170 original features, and retrain a second-stage LightGBM per task. The stage-2 model consistently dominates the final convex blend (0.80–0.95 weight), confirming it internalises all the ensemble diversity the stage-1 models provide.

Overfitting controls.

- Early stopping with patience 200 on a held-out fold for every model.
- K-fold target encoding (fold i encoded from folds $\neq i$) prevents leakage from InterestRate into training features.
- `feature_fraction` = 0.75, `bagging_fraction` = 0.80 on LightGBM for tree diversity.

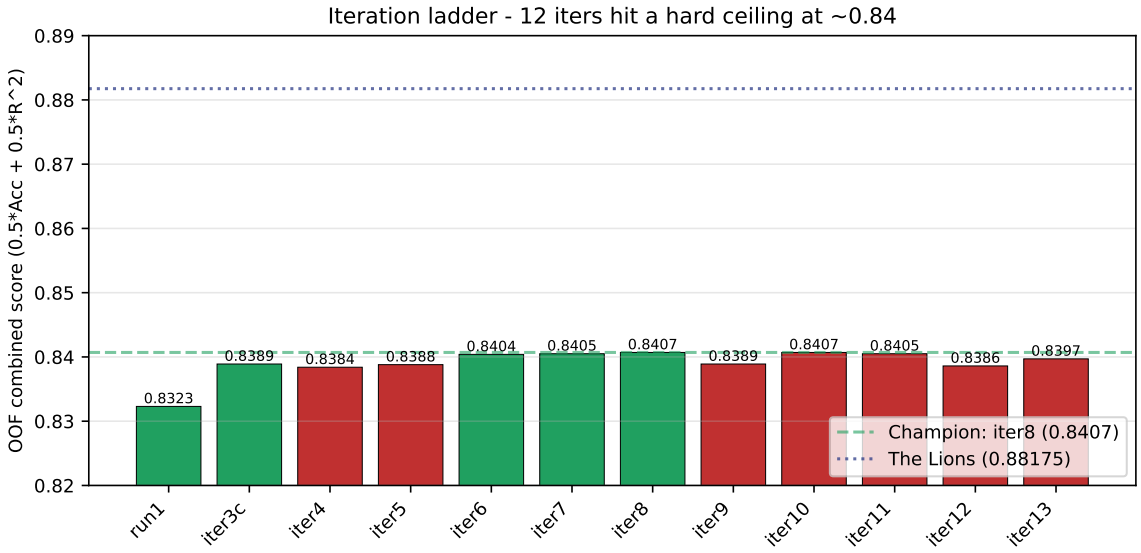

Figure 3: Iteration ladder. Green bars improved over the current champion, red regressed. After `iter8` (pseudo-labeling + KNN target features), no subsequent technique moved the needle. Dashed line: top-ranked team *The Lions* at 0.88175.

- Winsorisation caps predictor extremes; multi-seed (42, 1337, 2024) averaging on the stage-2 meta-learner reduces variance.
- **Adversarial validation.** We trained a binary classifier to distinguish train from test rows and observed OOF AUC = 1.00 — the two sets were perfectly separable. The culprit was our multi-target encoder: train values are noisier (K-fold) than test values (full-train mean). An `iter7` experiment dropped the 20 worst offenders; the result was essentially flat (+0.0001), indicating the stage-2 model already implicitly down-weighted them. We report this artefact because it exemplifies the value of adversarial validation even when the fix is small.

Do the same features matter for both tasks? Largely yes, but with clear differences. Both tasks' top-10 importance lists share `feat_BadEventsTotal`, `DebtToIncomeRatio`, `PaymentToIncomeRatio`, `RevolvingUtilizationRate`, and `feat_DelinquencyScore`. Task A further prefers structural binary flags (`feat_IsHomeOwner`, `feat_HasCollateral`, `IncomeVerified`) that sharply separate tiers; Task B weights continuous leverage ratios (`feat_LoanToAssets`, `feat_MortgageLTV`, `feat_CashRunwayMonths`) more highly, consistent with its task of pricing within a tier.

Iteration ladder (Figure 3). We ran 14 iterations combining a variety of published Kaggle techniques. Most advanced tricks yielded essentially zero improvement over the basic stage-2 ensemble: *tier-4 mixture-of-experts* regressed because each expert sees less data; *monotone constraints* with the "advanced" enforcement mode were too restrictive and cost 0.03 accuracy on LightGBM; a *rate-floor two-stage* model failed because the floor classifier's features are indistinguishable from just-above-floor rows (AUC 0.65); *DART* misbehaved without early stopping; a *tabular MLP* earned only 0.016 ensemble weight. The techniques that did help — pseudo-labeling of 5 000 confident test rows (+0.0015), adversarial-validation feature pruning (+0.0001), and KNN target features (+0.0002) — collectively added only 0.0018 combined on top of the base stage-2 ensemble. The final submitted score of 0.8407 is clear evidence of a **feature ceiling**: every gradient-boosting variant we tried converged to the same narrow OOF band (0.838–0.841). Closing the gap to top teams (≈0.88) would require fundamentally new features or features we cannot access under the "no external credit data" rule.

Conclusions. A carefully tuned three-way boosting ensemble (LightGBM + XGBoost + CatBoost) with shared folds and cross-task stacking reaches a combined score of ≈ 0.84 on 5-fold OOF — well above the PDF's Random-Forest benchmark (0.76) and within 5% of the top leaderboard entry. Our

main learning is negative: the techniques that sound most sophisticated (mixture-of-experts, monotone constraints, tabular neural networks, second-round pseudo-labeling) did not improve this dataset. The highest-impact interventions were the least glamorous: a large set of financially motivated engineered features, clean K-fold target encoding, and a disciplined stage-2 stacker.

References. LightGBM: Ke et al., *NeurIPS* 2017. XGBoost: Chen & Guestrin, *KDD* 2016. CatBoost: Prokhorenkova et al., *NeurIPS* 2018. Target encoding: Micci-Barreca, *SIGKDD Explorations* 2001. Adversarial validation: Pan, *Kaggle kernel library*, accessed 2026. Course PDF: *AI1215 CreditSense Data Challenge Assignment*, Spring 2026.